# LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

## Practice Pack: Sound

*Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow*

Name: _____ Roll No.: _____ Date: _____

**Marks obtained:** _____

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**General instructions:**

1. Level 1 checks the ideas, Level 2 is board-style numericals, and Level 3 is Foundation-level challenge.
2. Key relations: v = fλ (speed = frequency × wavelength) · f = 1/T · echo distance = (speed × time)/2

### LEVEL 1: CONCEPTS

**Q1.** Why can sound not travel through a vacuum?

**Q2.** Define frequency and give its SI unit.

**Q3.** What is the audible range of frequencies for human beings?

**Q4.** Which property of sound decides its (a) loudness and (b) pitch?

**Q5.** Why are the ceilings of concert halls and auditoriums curved?

### LEVEL 2: NUMERICALS

**Q6.** A sound wave has a frequency of 2 kHz and a wavelength of 35 cm. How long will it take to travel 1.5 km?

**Q7.** A person claps near a cliff and hears the echo after 2 s. How far is the cliff? (Speed of sound = 346 m/s)

**Q8.** A source vibrates at 100 Hz. How many times does it vibrate in one minute?

**Q9.** A sonar on a submarine receives the echo of its signal 5 s after sending it. The object is 3625 m away. Find the speed of sound in water.

**Q10.** Find the wavelength of a 220 Hz sound in air, where its speed is 440 m/s.

## LEVEL 3: CHALLENGE (FOUNDATION)

**Q11.** A stone is dropped from the top of a 500 m tall tower into a pond at its base. After how long is the splash heard at the top? (g = 10 $m/s^2$, speed of sound = 340 m/s)

**Q12.** What is the minimum distance needed between you and a wall to hear a distinct echo? (The sensation of a sound lasts about 0.1 s; speed of sound = 344 m/s.)

**Q13.** Bats fly in the dark without hitting objects. Explain how they do it.

*~ Echo questions: remember to halve the total distance! ~*